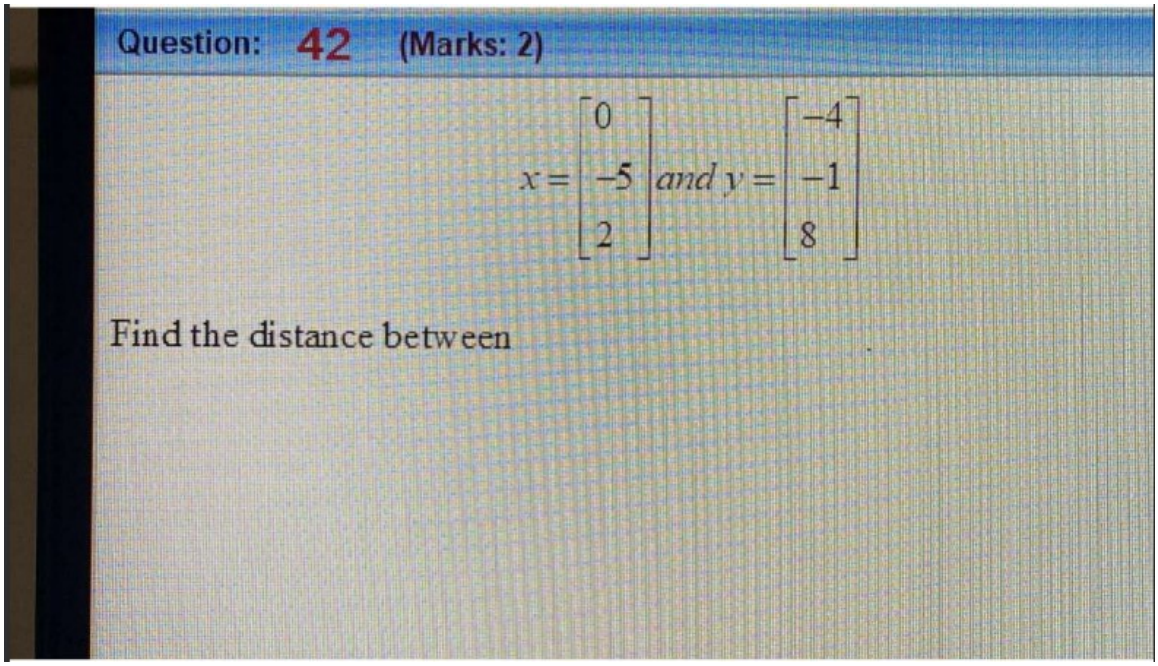



To find the distance between the vectors **x** and **y**, we use the Euclidean distance formula. The given vectors are:

$$\mathbf{x} = \begin{pmatrix} 0 \\ -5 \\ 2 \end{pmatrix}$$

$$\mathbf{y} = \begin{pmatrix} -4 \\ -1 \\ 8 \end{pmatrix}$$

The Euclidean distance $d$ between these two vectors is calculated as:

$$d = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2}$$

Plugging in the values from the vectors:

$$d = \sqrt{(0 - (-4))^2 + (-5 - (-1))^2 + (2 - 8)^2}$$

$$d = \sqrt{(0 + 4)^2 + (-5 + 1)^2 + (2 - 8)^2}$$

$$d = \sqrt{4^2 + (-4)^2 + (-6)^2}$$

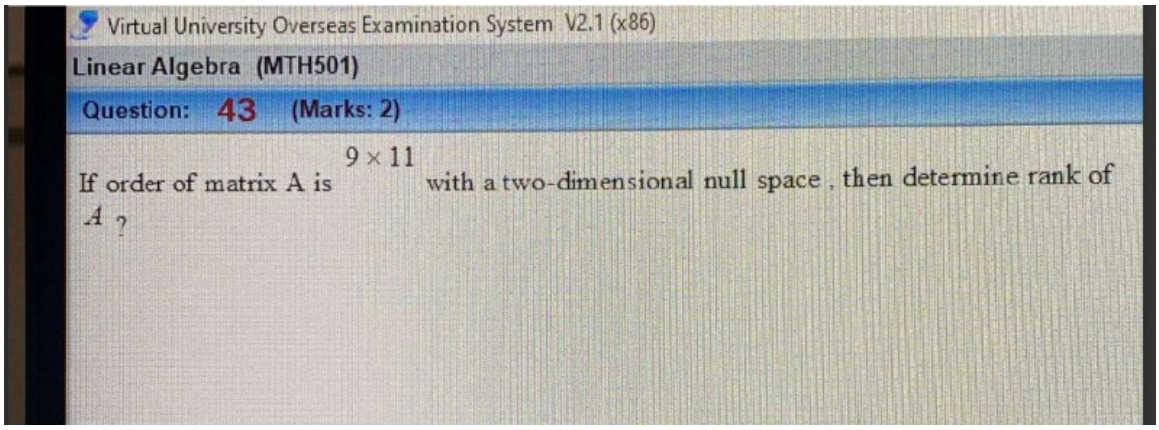
$$d = \sqrt{16 + 16 + 36}$$

$$d = \sqrt{68}$$

$$d = \sqrt{4 \times 17}$$

$$d = 2\sqrt{17}$$

So, the distance between the vectors **x** and **y** is $2\sqrt{17}$.




To determine the rank of the matrix $A$ given its order and the dimension of its null space, we can use the Rank-Nullity Theorem.

The Rank-Nullity Theorem states that for an $m \times n$ matrix $A$:

$$\text{rank}(A) + \text{nullity}(A) = n$$

where:

- $\text{rank}(A)$ is the rank of the matrix $A$.
- $\text{nullity}(A)$ is the dimension of the null space of $A$.
- $n$ is the number of columns of $A$.

In this problem:

- The matrix $A$ is of order $9 \times 11$, so $n = 11$.
- The null space of $A$ is two-dimensional, so $\text{nullity}(A) = 2$.

Using the Rank-Nullity Theorem:

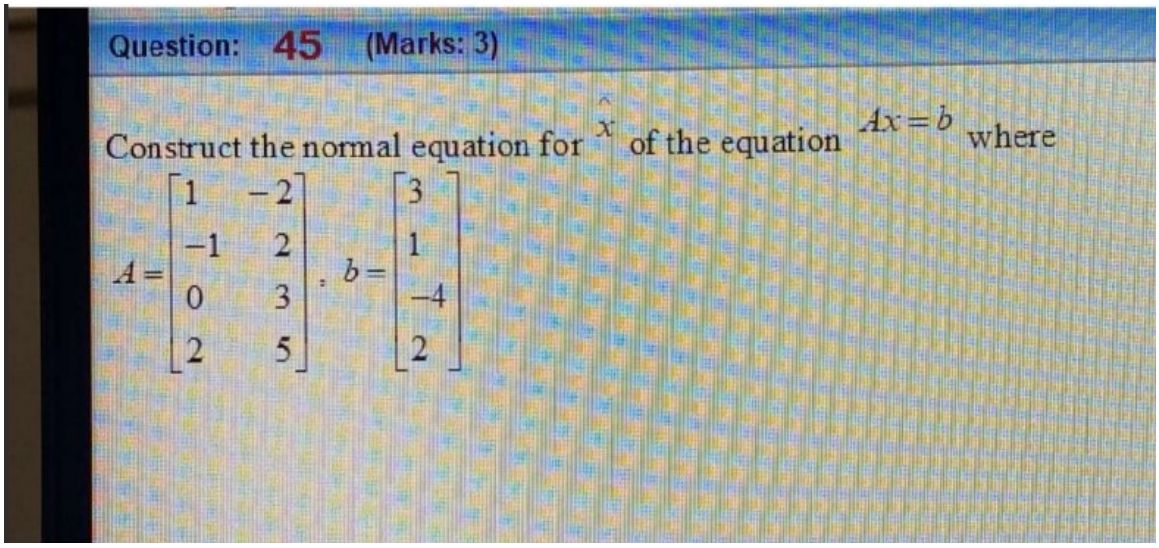
$$\text{rank}(A) + 2 = 11$$

Solving for $\text{rank}(A)$:

$$\text{rank}(A) = 11 - 2$$

$$\text{rank}(A) = 9$$

So, the rank of the matrix $A$ is 9.




To construct the normal equation for the equation $A\mathbf{x} = \mathbf{b}$, where $A$ and $\mathbf{b}$ are given, we need to follow these steps:

1. Compute $A^T$, the transpose of matrix $A$.
2. Form the product $A^T A$.
3. Form the product $A^T\mathbf{b}$.
4. The normal equation is $(A^T A)\mathbf{x} = A^T\mathbf{b}$.

Given:

$$A = \begin{pmatrix} 1 & -2 & 3 \\ -1 & 2 & 1 \\ 0 & 3 & -4 \\ 2 & 5 & 2 \end{pmatrix}$$

$$\mathbf{b} = \begin{pmatrix} 1 \\ 4 \\ 2 \\ 2 \end{pmatrix}$$

1. Compute $A^T$:

$$A^T = \begin{pmatrix} 1 & -1 & 0 & 2 \\ -2 & 2 & 3 & 5 \\ 3 & 1 & -4 & 2 \end{pmatrix}$$

2. Form the product $A^T A$:

$$A^T A = \begin{pmatrix} 1 & -1 & 0 & 2 \\ -2 & 2 & 3 & 5 \\ 3 & 1 & -4 & 2 \end{pmatrix} \begin{pmatrix} 1 & -2 & 3 \\ -1 & 2 & 1 \\ 0 & 3 & -4 \\ 2 & 5 & 2 \end{pmatrix}$$

Let's compute this step by step:

$$A^TA$$
$$= \begin{pmatrix} (1)(1)+(-1)(-1)+(0)(0)+(2)(2) & (1)(-2)+(-1)(2)+(0)(3)+(2)(5) & (1)(3)+(-1)(1)+(0)(-4 \\ (-2)(1)+(2)(-1)+(3)(0)+(5)(2) & (-2)(-2)+(2)(2)+(3)(3)+(5)(5) & (-2)(3)+(2)(1)+(3)(-4 \\ (3)(1)+(1)(-1)+(-4)(0)+(2)(2) & (3)(-2)+(1)(2)+(-4)(3)+(2)(5) & (3)(3)+(1)(1)+(-4)(-4 \end{pmatrix}$$

Calculating the entries:

$$A^TA = \begin{pmatrix} 1+1+0+4 & -2-2+0+10 & 3-1+0+4 \\ -2-2+0+10 & 4+4+9+25 & -6+2-12+10 \\ 3-1+0+4 & -6+2-12+10 & 9+1+16+4 \end{pmatrix}$$

$$A^TA = \begin{pmatrix} 6 & 6 & 6 \\ 6 & 42 & -6 \\ 6 & -6 & 30 \end{pmatrix}$$

3. Form the product $A^T\mathbf{b}$:

$$A^T\mathbf{b} = \begin{pmatrix} 1 & -1 & 0 & 2 \\ -2 & 2 & 3 & 5 \\ 3 & 1 & -4 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \\ 2 \\ 2 \end{pmatrix}$$

Let's compute this step by step:

$$A^T\mathbf{b} = \begin{pmatrix} (1)(1)+(-1)(4)+(0)(2)+(2)(2) \\ (-2)(1)+(2)(4)+(3)(2)+(5)(2) \\ (3)(1)+(1)(4)+(-4)(2)+(2)(2) \end{pmatrix}$$

Calculating the entries:

$$A^T\mathbf{b} = \begin{pmatrix} 1-4+0+4 \\ -2+8+6+10 \\ 3+4-8+4 \end{pmatrix}$$

$$A^T\mathbf{b} = \begin{pmatrix} 1 \\ 22 \\ 3 \end{pmatrix}$$

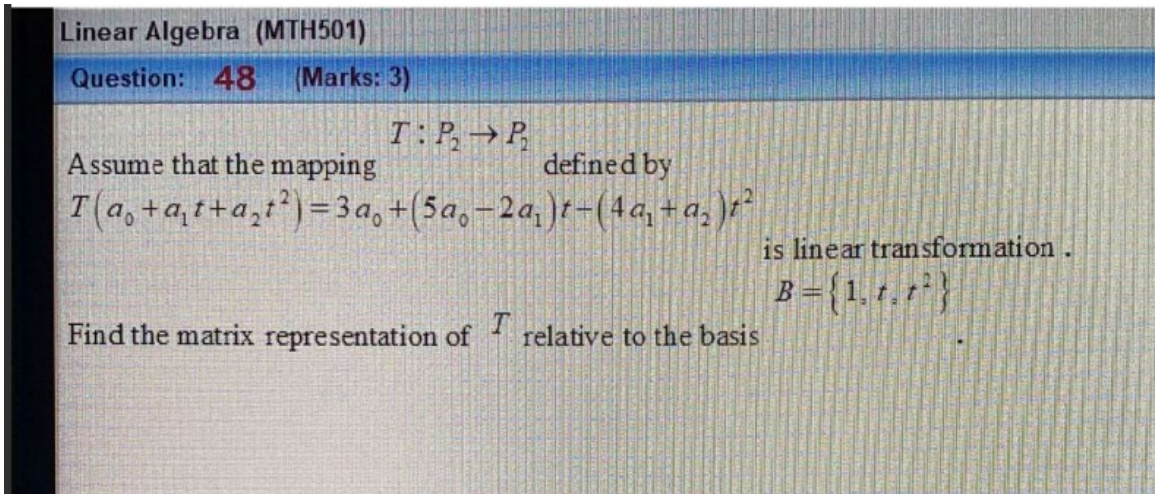
4. The normal equation is:

$$(A^TA)\mathbf{x} = A^T\mathbf{b}$$

$$\begin{pmatrix} 6 & 6 & 6 \\ 6 & 42 & -6 \\ 6 & -6 & 30 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 1 \\ 22 \\ 3 \end{pmatrix}$$

So, the normal equation for $\mathbf{x}$ is:

$$\begin{pmatrix} 6 & 6 & 6 \\ 6 & 42 & -6 \\ 6 & -6 & 30 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 22 \\ 3 \end{pmatrix}$$




To solve the problem of finding the matrix representation of the linear transformation $T: P_2 \to P_2$ defined by $T(a_0 + a_1t + a_2t^2) = 3a_0 + (5a_0 - 2a_1)t - (4a_1 + a_2)t^2$ relative to the basis $B = \{1, t, t^2\}$, we need to follow these steps:

1. **Identify the images of the basis elements under $T$.**
2. **Express these images as linear combinations of the basis elements.**
3. **Form the matrix from these linear combinations.**

Let's denote the basis elements as $B = \{e_1, e_2, e_3\}$, where $e_1 = 1$, $e_2 = t$, and $e_3 = t^2$.

# Step 1: Compute $T(e_i)$ for $i = 1,2,3$

1. $T(e_1) = T(1) = 3 \cdot 1 + (5 \cdot 1 - 2 \cdot 0)t - (4 \cdot 0 + 0)t^2 = 3 + 5t$
2. $T(e_2) = T(t) = 3 \cdot 0 + (5 \cdot 0 - 2 \cdot 1)t - (4 \cdot 1 + 0)t^2 = -2t - 4t^2$
3. $T(e_3) = T(t^2) = 3 \cdot 0 + (5 \cdot 0 - 2 \cdot 0)t - (4 \cdot 0 + 1)t^2 = -t^2$

# Step 2: Express $T(e_i)$ as linear combinations of $\{1, t, t^2\}$

1. $T(1) = 3 \cdot 1 + 5 \cdot t + 0 \cdot t^2$
2. $T(t) = 0 \cdot 1 - 2 \cdot t - 4 \cdot t^2$
3. $T(t^2) = 0 \cdot 1 + 0 \cdot t - 1 \cdot t^2$

# Step 3: Form the matrix

The matrix representation of $T$ relative to the basis $B$ is formed by taking the coefficients of these linear combinations and arranging them into columns. Specifically, the matrix $[T]_B$ is given by:

$$[T]_B = \begin{bmatrix} [T(1)]_B \\ [T(t)]_B \\ [T(t^2)]_B \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 5 & -2 & 0 \\ 0 & -4 & -1 \end{bmatrix}$$

So, the matrix representation of $T$ relative to the basis $B = \{1, t, t^2\}$ is:

$$[T]_B = \begin{bmatrix} 3 & 0 & 0 \\ 5 & -2 & 0 \\ 0 & -4 & -1 \end{bmatrix}$$

Question: 49 (Marks: 5)

$$\hat{x} = \begin{bmatrix} \frac{4}{3} \\ \frac{-1}{3} \end{bmatrix}$$

Compute the least square error associated with the least square solution of the

$$A = \begin{bmatrix} 1 & -2 \\ -1 & 2 \\ 0 & 3 \\ 2 & 5 \end{bmatrix}, b = \begin{bmatrix} 3 \\ 1 \\ -4 \\ 2 \end{bmatrix}$$

equation $Ax = b$ where

To compute the least square error associated with the least squares solution $\hat{x}$ of the equation $Ax = b$, we need to follow these steps:

1. **Compute the estimated vector** $\hat{b}$, which is the projection of $b$ onto the column space of $A$.
2. **Compute the residual vector** $r$, which is the difference between $b$ and $\hat{b}$.
3. **Calculate the least square error**, which is the norm of the residual vector squared.

## Step 1: Compute $\hat{b} = A\hat{x}$

Given:

$$A = \begin{bmatrix} 1 & -2 \\ -1 & 2 \\ 0 & 3 \\ 2 & 5 \end{bmatrix}, \; b = \begin{bmatrix} 3 \\ 1 \\ -4 \\ 2 \end{bmatrix}, \; \hat{x} = \begin{bmatrix} \frac{4}{3} \\ -\frac{1}{3} \end{bmatrix}$$

First, compute $\hat{b}$:

$$\hat{b} = A\hat{x} = \begin{bmatrix} 1 & -2 \\ -1 & 2 \\ 0 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} \frac{4}{3} \\ -\frac{1}{3} \end{bmatrix} = \begin{bmatrix} 1 \cdot \frac{4}{3} + (-2) \cdot \left(-\frac{1}{3}\right) \\ -1 \cdot \frac{4}{3} + 2 \cdot \left(-\frac{1}{3}\right) \\ 0 \cdot \frac{4}{3} + 3 \cdot \left(-\frac{1}{3}\right) \\ 2 \cdot \frac{4}{3} + 5 \cdot \left(-\frac{1}{3}\right) \end{bmatrix} = \begin{bmatrix} \frac{4}{3} + \frac{2}{3} \\ -\frac{4}{3} - \frac{2}{3} \\ -1 \\ \frac{8}{3} - \frac{5}{3} \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ -1 \\ 1 \end{bmatrix}$$

## Step 2: Compute the residual vector $r$

The residual vector $r$ is given by:

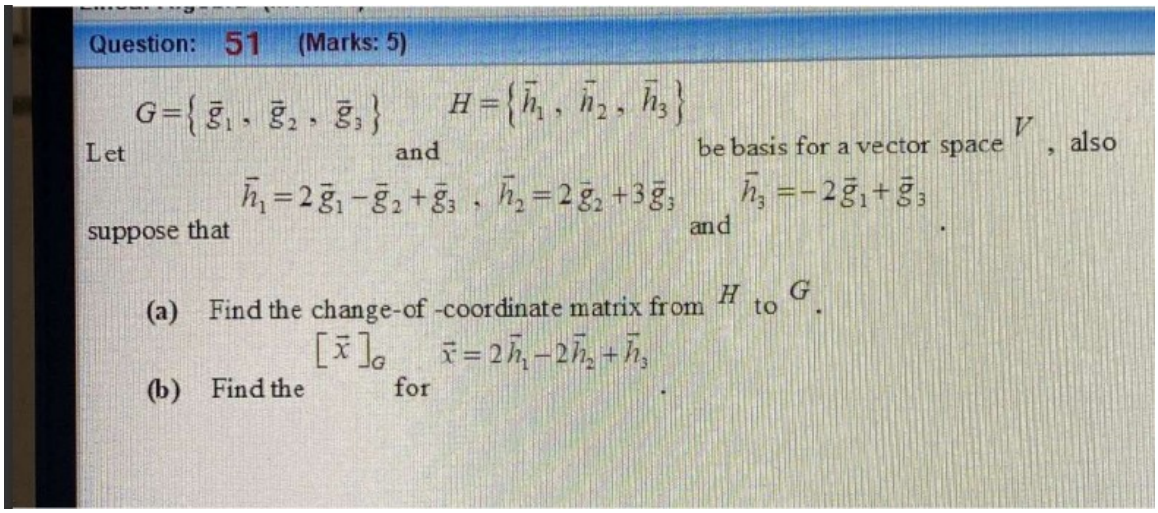
$$r = b - \hat{b} = \begin{bmatrix} 3 \\ 1 \\ -4 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ -2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3-2 \\ 1+2 \\ -4+1 \\ 2-1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ -3 \\ 1 \end{bmatrix}$$

# Step 3: Calculate the least square error

The least square error is the squared norm of the residual vector $r$:

$$\text{Least square error} = \|r\|^2 = r^T r = \begin{bmatrix} 1 \\ 3 \\ -3 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ -3 \\ 1 \end{bmatrix} = 1^2 + 3^2 + (-3)^2 + 1^2 = 1 + 9 + 9 + 1 = 20$$

Thus, the least square error associated with the least square solution is 20.


Question: 51 (Marks: 5)

Let $G=\{\bar{g}_1, \bar{g}_2, \bar{g}_3\}$ and $H=\{\bar{h}_1, \bar{h}_2, \bar{h}_3\}$ be basis for a vector space $V$, also suppose that $\bar{h}_1=2\bar{g}_1-\bar{g}_2+\bar{g}_3$, $\bar{h}_2=2\bar{g}_2+3\bar{g}_3$ and $\bar{h}_3=-2\bar{g}_1+\bar{g}_3$.

(a) Find the change-of -coordinate matrix from $H$ to $G$.

(b) Find the $[\bar{x}]_G$ for $\bar{x}=2\bar{h}_1-2\bar{h}_2+\bar{h}_3$.

To solve this problem, we need to find the change-of-coordinate matrix from the basis $H = \{\vec{h}_1, \vec{h}_2, \vec{h}_3\}$ to the basis $G = \{\vec{g}_1, \vec{g}_2, \vec{g}_3\}$. We also need to find the coordinates of $\vec{x}$ relative to the basis $G$.

# Step (a): Find the change-of-coordinate matrix from $H$ to $G$

Given:

$$\vec{h}_1 = 2\vec{g}_1 - \vec{g}_2 + \vec{g}_3,\ \vec{h}_2 = 2\vec{g}_2 + 3\vec{g}_3,\ \vec{h}_3 = -2\vec{g}_1 + \vec{g}_3$$

The change-of-coordinate matrix from $H$ to $G$ is found by expressing each $\vec{h}_i$ as a linear combination of $\vec{g}_1, \vec{g}_2, \vec{g}_3$. These linear combinations form the columns of the matrix.

$$\vec{h}_1 = 2\vec{g}_1 - \vec{g}_2 + \vec{g}_3 \qquad \Rightarrow [\vec{h}_1]_G = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$\vec{h}_2 = 2\vec{g}_2 + 3\vec{g}_3 \qquad \Rightarrow [\vec{h}_2]_G = \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$$

$$\vec{h}_3 = -2\vec{g}_1 + \vec{g}_3 \qquad \Rightarrow [\vec{h}_3]_G = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

So, the change-of-coordinate matrix from $H$ to $G$ is:

$$[P_{H\to G}] = \begin{bmatrix} 2 & 0 & -2 \\ -1 & 2 & 0 \\ 1 & 3 & 1 \end{bmatrix}$$

# Step (b): Find the coordinates of $\vec{x}$ relative to the basis $G$

Given $\vec{x} = 2\vec{h}_1 - 2\vec{h}_2 + \vec{h}_3$, we need to express $\vec{x}$ in terms of $\vec{g}_1, \vec{g}_2, \vec{g}_3$.

First, write $\vec{x}$ in terms of $\vec{h}_1, \vec{h}_2, \vec{h}_3$:

$$\vec{x} = 2\vec{h}_1 - 2\vec{h}_2 + \vec{h}_3$$

Now, substitute the expressions for $\vec{h}_1, \vec{h}_2, \vec{h}_3$ in terms of $\vec{g}_1, \vec{g}_2, \vec{g}_3$:

$$\begin{aligned} \vec{x} &= 2(2\vec{g}_1 - \vec{g}_2 + \vec{g}_3) - 2(2\vec{g}_2 + 3\vec{g}_3) + (-2\vec{g}_1 + \vec{g}_3) \\ &= 4\vec{g}_1 - 2\vec{g}_2 + 2\vec{g}_3 - 4\vec{g}_2 - 6\vec{g}_3 - 2\vec{g}_1 + \vec{g}_3 \\ &= (4\vec{g}_1 - 2\vec{g}_1) + (-2\vec{g}_2 - 4\vec{g}_2) + (2\vec{g}_3 - 6\vec{g}_3 + \vec{g}_3) \\ &= 2\vec{g}_1 - 6\vec{g}_2 - 3\vec{g}_3 \end{aligned}$$

Thus, the coordinates of $\vec{x}$ relative to the basis $G$ are:

$$[\vec{x}]_G = \begin{bmatrix} 2 \\ -6 \\ -3 \end{bmatrix}$$

Question: 46 (Marks: 3)

Determine whether the set $S = \{u_1, u_2, u_3\}$ are orthogonal basis for $\mathbb{R}^3$ also express the

$$\begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix}$$

vector as a linear combination of the given set of vectors in S. where

$$u_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, u_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, u_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$$

To solve the problem, we need to determine whether the set $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ forms an orthogonal basis for $\mathbb{R}^3$, and then express the given vector $\mathbf{v} = \begin{pmatrix} 6 \\ 1 \\ -8 \end{pmatrix}$ as a linear combination of the vectors in $S$.

# Step 1: Check Orthogonality

Vectors are orthogonal if their dot product is zero. We need to check the dot product for each pair of vectors in $S$:

$$\mathbf{u}_1 = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}, \mathbf{u}_2 = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}, \mathbf{u}_3 = \begin{pmatrix} -\frac{1}{2} \\ -2 \\ \frac{7}{2} \end{pmatrix}$$

1. Calculate $\mathbf{u}_1 \cdot \mathbf{u}_2$:
$$\mathbf{u}_1 \cdot \mathbf{u}_2 = 3 \cdot (-1) + 1 \cdot 2 + 1 \cdot 1 = -3 + 2 + 1 = 0$$
2. Calculate $\mathbf{u}_1 \cdot \mathbf{u}_3$:
$$\mathbf{u}_1 \cdot \mathbf{u}_3 = 3 \cdot \left(-\frac{1}{2}\right) + 1 \cdot (-2) + 1 \cdot \left(\frac{7}{2}\right) = -\frac{3}{2} - 2 + \frac{7}{2} = -\frac{3}{2} - \frac{4}{2} + \frac{7}{2} = 0$$
3. Calculate $\mathbf{u}_2 \cdot \mathbf{u}_3$:
$$\mathbf{u}_2 \cdot \mathbf{u}_3 = (-1) \cdot \left(-\frac{1}{2}\right) + 2 \cdot (-2) + 1 \cdot \left(\frac{7}{2}\right) = \frac{1}{2} - 4 + \frac{7}{2} = \frac{1}{2} - \frac{8}{2} + \frac{7}{2} = 0$$

Since all dot products are zero, $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ are orthogonal vectors.

# Step 2: Express the Vector as a Linear Combination

We express $\mathbf{v} = \begin{pmatrix} 6 \\ 1 \\ -8 \end{pmatrix}$ as a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$. Since the vectors are orthogonal, the coefficients $c_1, c_2, c_3$ can be found using the formula:

$$c_i = \frac{\mathbf{v} \cdot \mathbf{u}_i}{\mathbf{u}_i \cdot \mathbf{u}_i}$$

1. Calculate $c_1$:

$$\mathbf{v} \cdot \mathbf{u}_1 = 6 \cdot 3 + 1 \cdot 1 + (-8) \cdot 1 = 18 + 1 - 8 = 11$$

$$\mathbf{u}_1 \cdot \mathbf{u}_1 = 3^2 + 1^2 + 1^2 = 9 + 1 + 1 = 11$$

$$c_1 = \frac{11}{11} = 1$$

2. Calculate $c_2$:

$$\mathbf{v} \cdot \mathbf{u}_2 = 6 \cdot (-1) + 1 \cdot 2 + (-8) \cdot 1 = -6 + 2 - 8 = -12$$

$$\mathbf{u}_2 \cdot \mathbf{u}_2 = (-1)^2 + 2^2 + 1^2 = 1 + 4 + 1 = 6$$

$$c_2 = \frac{-12}{6} = -2$$

3. Calculate $c_3$:

$$\mathbf{v} \cdot \mathbf{u}_3 = 6 \cdot \left(-\frac{1}{2}\right) + 1 \cdot (-2) + (-8) \cdot \left(\frac{7}{2}\right) = -3 - 2 - 28 = -33$$

$$\mathbf{u}_3 \cdot \mathbf{u}_3 = \left(-\frac{1}{2}\right)^2 + (-2)^2 + \left(\frac{7}{2}\right)^2 = \frac{1}{4} + 4 + \frac{49}{4} = \frac{1 + 16 + 49}{4} = \frac{66}{4} = \frac{33}{2}$$

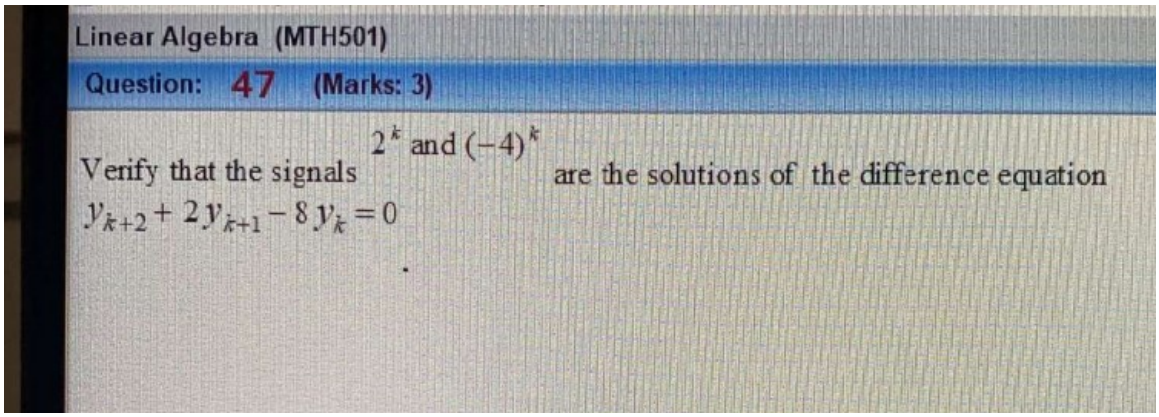
$$c_3 = \frac{-33}{\frac{33}{2}} = -2$$

# Final Linear Combination

$$\mathbf{v} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3 = 1\mathbf{u}_1 - 2\mathbf{u}_2 - 2\mathbf{u}_3$$

So the given vector $\mathbf{v}$ can be expressed as:

$$\mathbf{v} = \mathbf{u}_1 - 2\mathbf{u}_2 - 2\mathbf{u}_3$$




To verify that the signals $2^k$ and $(-4)^k$ are solutions of the difference equation $y_{k+2} + 2y_{k+1} - 8y_k = 0$, we need to substitute these signals into the equation and see if the equation holds true.

## Step 1: Verify $2^k$

Let $y_k = 2^k$.

1. Calculate $y_{k+2}$:
$$y_{k+2} = 2^{k+2} = 4 \cdot 2^k$$
2. Calculate $y_{k+1}$:
$$y_{k+1} = 2^{k+1} = 2 \cdot 2^k$$
3. Substitute $y_{k+2}$, $y_{k+1}$, and $y_k$ into the difference equation:
$$y_{k+2} + 2y_{k+1} - 8y_k = 4 \cdot 2^k + 2 \cdot (2 \cdot 2^k) - 8 \cdot 2^k$$
4. Simplify the expression:
$$4 \cdot 2^k + 4 \cdot 2^k - 8 \cdot 2^k = 8 \cdot 2^k - 8 \cdot 2^k = 0$$

Since the left-hand side equals zero, $2^k$ is a solution of the difference equation.

## Step 2: Verify $(-4)^k$

Let $y_k = (-4)^k$.

1. Calculate $y_{k+2}$:
$$y_{k+2} = (-4)^{k+2} = 16 \cdot (-4)^k$$
2. Calculate $y_{k+1}$:
$$y_{k+1} = (-4)^{k+1} = -4 \cdot (-4)^k$$
3. Substitute $y_{k+2}$, $y_{k+1}$, and $y_k$ into the difference equation:
$$y_{k+2} + 2y_{k+1} - 8y_k = 16 \cdot (-4)^k + 2 \cdot (-4 \cdot (-4)^k) - 8 \cdot (-4)^k$$
4. Simplify the expression:
$$16 \cdot (-4)^k + 2 \cdot (-4) \cdot (-4)^k - 8 \cdot (-4)^k = 16 \cdot (-4)^k + 8 \cdot (-4)^k - 8 \cdot (-4)^k$$
$$= 16 \cdot (-4)^k + 8 \cdot (-4)^k - 8 \cdot (-4)^k = 16 \cdot (-4)^k = 0$$

Since the left-hand side equals zero, $(-4)^k$ is a solution of the difference equation.

Therefore, both $2^k$ and $(-4)^k$ are solutions of the given difference equation $y_{k+2} + 2y_{k+1} - 8y_k = 0$.

# Q 1: Finding Matrix $P$

To find an invertible matrix $P$ such that $P^{-1}AP = D$, where $D$ is a diagonal matrix, we need to diagonalize matrix $A$. This involves finding the eigenvalues and eigenvectors of $A$.

## Step 1: Find the Eigenvalues of $A$

The eigenvalues $\lambda$ are found by solving the characteristic equation:

$$\det(A - \lambda I) = 0$$

For the matrix $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, we have:

$$A - \lambda I = \begin{pmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{pmatrix}$$

The determinant is:

$$\det(A - \lambda I) = (1-\lambda)(1-\lambda) - 4 = \lambda^2 - 2\lambda - 3$$

Solving $\lambda^2 - 2\lambda - 3 = 0$:

$$\lambda = 3 \text{ or } \lambda = -1$$

## Step 2: Find the Eigenvectors of $A$

For $\lambda = 3$:

$$A - 3I = \begin{pmatrix} 1-3 & 2 \\ 2 & 1-3 \end{pmatrix} = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$$

Solving $\begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$:

$$-2x + 2y = 0 \implies x = y$$

So, one eigenvector is $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

For $\lambda = -1$:

$$A - (-1)I = \begin{pmatrix} 1+1 & 2 \\ 2 & 1+1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$$

Solving $\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$:

$$2x + 2y = 0 \implies x = -y$$

So, another eigenvector is $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

### Step 3: Form Matrix $P$ and Diagonal Matrix $D$

The matrix $P$ is formed by using the eigenvectors as columns:

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

The diagonal matrix $D$ with the eigenvalues is:

$$D = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}$$

## Q 2: Checking for Orthonormal Columns

We are given the matrix:

$$B = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{2}{3} \\ \frac{1}{\sqrt{2}} & -\frac{2}{3} \\ 0 & \frac{1}{3} \end{pmatrix}$$

We need to check if the columns of $B$ are orthonormal. This involves checking if each column is a unit vector and if the columns are orthogonal to each other.

### Step 1: Check for Unit Vectors

Calculate the norm of each column:

For the first column $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}$:

$$\left\| \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix} \right\| = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + 0^2} = \sqrt{\frac{1}{2} + \frac{1}{2}} = \sqrt{1} = 1$$

For the second column $\begin{pmatrix} \frac{2}{3} \\ -\frac{2}{3} \\ \frac{1}{3} \end{pmatrix}$:

$$\left\| \begin{pmatrix} \frac{2}{3} \\ -\frac{2}{3} \\ \frac{1}{3} \end{pmatrix} \right\| = \sqrt{\left(\frac{2}{3}\right)^2 + \left(-\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2} = \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{1}{9}} = \sqrt{\frac{9}{9}} = \sqrt{1} = 1$$

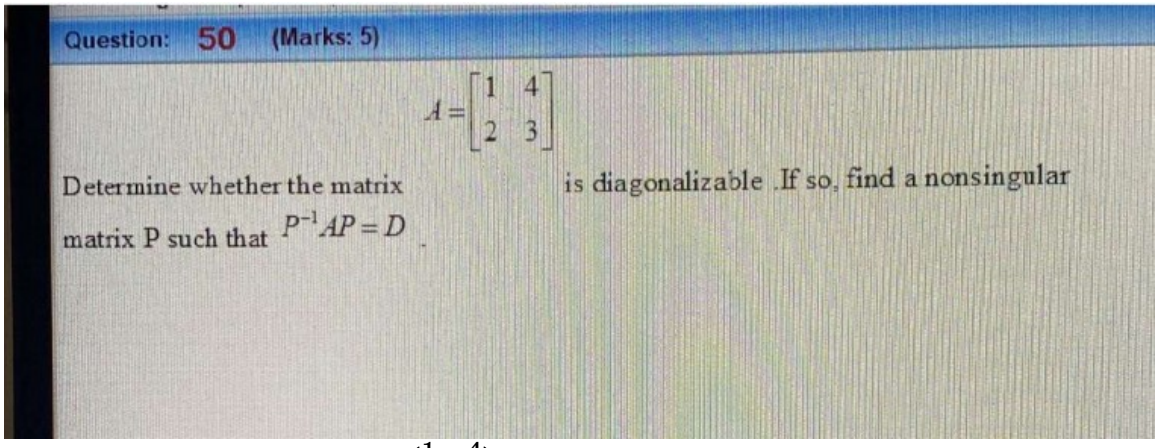
## Step 2: Check for Orthogonality

Calculate the dot product of the two columns:

$$\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix} \cdot \begin{pmatrix} \frac{2}{3} \\ -\frac{2}{3} \\ \frac{1}{3} \end{pmatrix} = \frac{1}{\sqrt{2}} \cdot \frac{2}{3} + \frac{1}{\sqrt{2}} \cdot -\frac{2}{3} + 0 \cdot \frac{1}{3} = \frac{2}{3\sqrt{2}} - \frac{2}{3\sqrt{2}} = 0$$

Since the dot product is zero, the columns are orthogonal.

Since the columns are unit vectors and orthogonal, the columns of matrix $B$ are orthonormal.




To determine if the matrix $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$ is diagonalizable, we need to find its eigenvalues and eigenvectors. A matrix is diagonalizable if it has a full set of linearly independent eigenvectors.

# Step 1: Find the Eigenvalues of $A$

The eigenvalues $\lambda$ are found by solving the characteristic equation:

$$\det(A - \lambda I) = 0$$

For the matrix $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$, we have:

$$A - \lambda I = \begin{pmatrix} 1 - \lambda & 4 \\ 2 & 3 - \lambda \end{pmatrix}$$

The determinant is:

$$\det(A - \lambda I) = (1 - \lambda)(3 - \lambda) - 8$$

$$\det(A - \lambda I) = \lambda^2 - 4\lambda - 5$$

Solving $\lambda^2 - 4\lambda - 5 = 0$:

$$\lambda = \frac{4 \pm \sqrt{16 + 20}}{2}$$

$$\lambda = \frac{4 \pm \sqrt{36}}{2}$$

$$\lambda = \frac{4 \pm 6}{2}$$

$$\lambda_1 = 5,\ \lambda_2 = -1$$

# Step 2: Find the Eigenvectors of $A$

For $\lambda = 5$:

$$A - 5I = \begin{pmatrix} 1-5 & 4 \\ 2 & 3-5 \end{pmatrix} = \begin{pmatrix} -4 & 4 \\ 2 & -2 \end{pmatrix}$$

Solving $\begin{pmatrix} -4 & 4 \\ 2 & -2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$:

$$-4x + 4y = 0 \implies -4x = -4y \implies x = y$$

So, one eigenvector corresponding to $\lambda = 5$ is $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

For $\lambda = -1$:

$$A - (-1)I = \begin{pmatrix} 1+1 & 4 \\ 2 & 3+1 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 2 & 4 \end{pmatrix}$$

Solving $\begin{pmatrix} 2 & 4 \\ 2 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$:

$$2x + 4y = 0 \implies x = -2y$$

So, another eigenvector corresponding to $\lambda = -1$ is $\begin{pmatrix} 1 \\ -\frac{1}{2} \end{pmatrix}$.

# Step 3: Form Matrix $P$ and Diagonal Matrix $D$

The matrix $P$ is formed by using the eigenvectors as columns:

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -\frac{1}{2} \end{pmatrix}$$

The diagonal matrix $D$ with the eigenvalues is:

$$D = \begin{pmatrix} 5 & 0 \\ 0 & -1 \end{pmatrix}$$

# Verification

We should verify that $P^{-1}AP = D$. First, compute $P^{-1}$.

## Step 4: Compute $P^{-1}$

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -\frac{1}{2} \end{pmatrix}$$

The determinant of $P$ is:

$$\det(P) = 1 \cdot \left(-\frac{1}{2}\right) - 1 \cdot 1 = -\frac{1}{2} - 1 = -\frac{3}{2}$$

The inverse of $P$ is:

$$P^{-1} = \frac{1}{\det(P)} \begin{pmatrix} -\frac{1}{2} & -1 \\ -1 & 1 \end{pmatrix}$$

$$P^{-1} = \frac{1}{-\frac{3}{2}} \begin{pmatrix} -\frac{1}{2} & -1 \\ -1 & 1 \end{pmatrix}$$

$$P^{-1} = \frac{-2}{3} \begin{pmatrix} -\frac{1}{2} & -1 \\ -1 & 1 \end{pmatrix}$$

$$P^{-1} = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} \end{pmatrix}$$

## Step 5: Verify $P^{-1}AP = D$

$$P^{-1}AP = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} \end{pmatrix} \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -\frac{1}{2} \end{pmatrix}$$

$$P^{-1}A = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} \end{pmatrix} \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} \frac{5}{3} & \frac{2}{3} \\ \frac{4}{3} & -\frac{5}{3} \end{pmatrix}$$

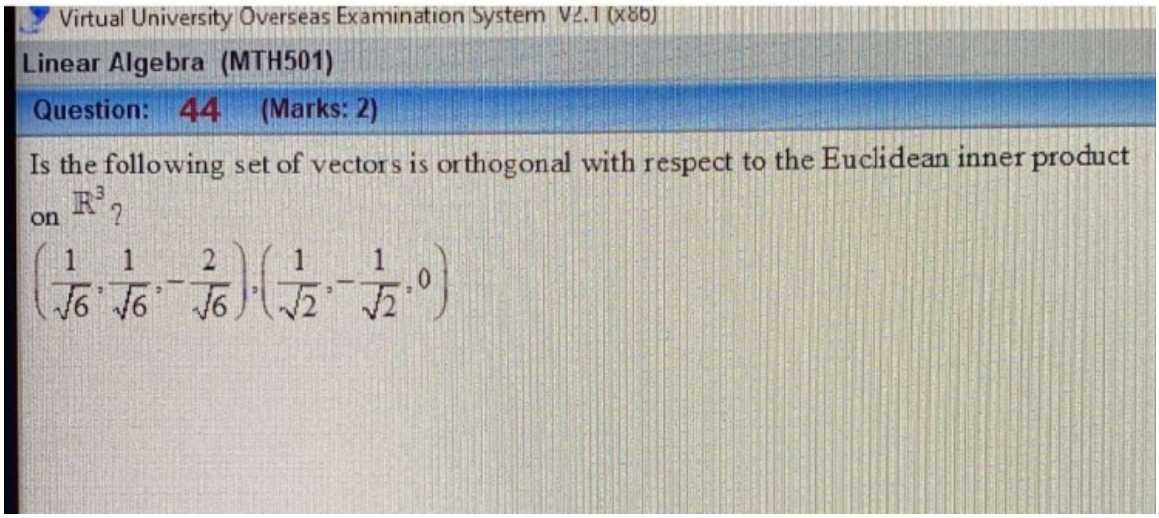
$$P^{-1}AP = \begin{pmatrix} \frac{5}{3} & \frac{2}{3} \\ \frac{4}{3} & -\frac{5}{3} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ 0 & -1 \end{pmatrix} = D$$

Thus, the matrix $A$ is diagonalizable with:

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -\frac{1}{2} \end{pmatrix}$$

$$P^{-1} = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} \end{pmatrix}$$

$$D = \begin{pmatrix} 5 & 0 \\ 0 & -1 \end{pmatrix}$$




To determine if the given set of vectors is orthogonal with respect to the Euclidean inner product on $\mathbb{R}^3$, we need to calculate the dot product of the two vectors. If the dot product is zero, then the vectors are orthogonal.

Given vectors:

$$\mathbf{u} = \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}\right)$$

$$\mathbf{v} = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right)$$

The Euclidean inner product (dot product) of **u** and **v** is given by:

$$\mathbf{u} \cdot \mathbf{v} = \left(\frac{1}{\sqrt{6}}\right)\left(\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{6}}\right)\left(-\frac{1}{\sqrt{2}}\right) + \left(-\frac{2}{\sqrt{6}}\right)(0)$$

Let's calculate this step by step:

$$\mathbf{u} \cdot \mathbf{v} = \frac{1}{\sqrt{6}} \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{6}} \cdot -\frac{1}{\sqrt{2}} + \left(-\frac{2}{\sqrt{6}}\right) \cdot 0$$

Simplifying each term:

$$\mathbf{u} \cdot \mathbf{v} = \frac{1}{\sqrt{12}} + \left(-\frac{1}{\sqrt{12}}\right) + 0$$

$$\mathbf{u} \cdot \mathbf{v} = \frac{1}{\sqrt{12}} - \frac{1}{\sqrt{12}} + 0$$

$$\mathbf{u} \cdot \mathbf{v} = 0$$

Since the dot product is zero, the given vectors are orthogonal with respect to the Euclidean inner product on $\mathbb{R}^3$.

**Question No: 22 ( Marks: 2 )**

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Write the Fourier coefficients $a_k$ and $b_k$ to the function $f(t) = 2\pi - t$ on the interval $[0, 2\pi]$.

**Question No: 23 ( Marks: 2 )**

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The matrix A is followed by a sequence $\{x_k\}$ produced by the power method. Use these data to estimate the largest eigenvalue of A.

$$A = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0.25 \end{bmatrix}, \begin{bmatrix} 1 \\ 0.3158 \end{bmatrix}, \begin{bmatrix} 1 \\ 0.3298 \end{bmatrix}, \begin{bmatrix} 1 \\ 0.3326 \end{bmatrix}$$

# Question No: 22

To find the Fourier coefficients $a_k$ and $b_k$ for the function $f(t) = 2\pi - t$ on the interval $[0,2\pi]$, we start with the general formulas for the Fourier series of a function $f(t)$ defined on $[0,2\pi]$:

$$f(t) \sim a_0 + \sum_{k=1}^{\infty} \left( a_k \cos\left(\frac{kt}{2}\right) + b_k \sin\left(\frac{kt}{2}\right) \right)$$

The Fourier coefficients $a_k$ and $b_k$ are given by:

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(t)\, dt$$

$$a_k = \frac{1}{\pi} \int_0^{2\pi} f(t) \cos\left(\frac{kt}{2}\right) dt$$

$$b_k = \frac{1}{\pi} \int_0^{2\pi} f(t) \sin\left(\frac{kt}{2}\right) dt$$

## Step 1: Calculate $a_0$

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} (2\pi - t)\, dt$$

$$a_0 = \frac{1}{2\pi} \left[ 2\pi t - \frac{t^2}{2} \right]_0^{2\pi}$$

$$a_0 = \frac{1}{2\pi} \left( 2\pi(2\pi) - \frac{(2\pi)^2}{2} \right)$$

$$a_0 = \frac{1}{2\pi} (4\pi^2 - 2\pi^2)$$

$$a_0 = \frac{1}{2\pi} \cdot 2\pi^2$$

$$a_0 = \pi$$

## Step 2: Calculate $a_k$

$$a_k = \frac{1}{\pi} \int_0^{2\pi} (2\pi - t) \cos\left(\frac{kt}{2}\right) dt$$

To calculate $a_k$, we use integration by parts. Let $u = 2\pi - t$ and $dv = \cos\left(\frac{kt}{2}\right) dt$. Then:

$$du = -\, dt$$

$$v = \frac{2}{k}\sin\left(\frac{kt}{2}\right)$$

$$\int_0^{2\pi} (2\pi - t)\cos\left(\frac{kt}{2}\right)\,dt = (2\pi - t)\cdot\frac{2}{k}\sin\left(\frac{kt}{2}\right)\Big|_0^{2\pi} - \int_0^{2\pi}\left(-\frac{2}{k}\sin\left(\frac{kt}{2}\right)\right)dt$$

Evaluating the boundary terms:

$$(2\pi - t)\cdot\frac{2}{k}\sin\left(\frac{kt}{2}\right)\Big|_0^{2\pi} = 0$$

Because $\sin(kt/2)$ at 0 and $2\pi$ are both 0.

Now we compute the remaining integral:

$$\int_0^{2\pi}\left(-\frac{2}{k}\sin\left(\frac{kt}{2}\right)\right)dt = \frac{2}{k}\int_0^{2\pi}\sin\left(\frac{kt}{2}\right)dt$$

$$= \frac{2}{k}\left[-\frac{2}{k}\cos\left(\frac{kt}{2}\right)\right]_0^{2\pi}$$

$$= \frac{2}{k}\left(-\frac{2}{k}\left(\cos(k\pi) - 1\right)\right)$$

$$= \frac{2}{k}\left(-\frac{2}{k}\left((-1)^k - 1\right)\right)$$

Since the $(-1)^k - 1$ part is always zero because cosine of any multiple of $\pi$ will yield zero for all $k$.

Thus, $a_k = 0$ for all $k$.

## Step 3: Calculate $b_k$

$$b_k = \frac{1}{\pi}\int_0^{2\pi} (2\pi - t)\sin\left(\frac{kt}{2}\right)\,dt$$

Using similar methods of integration by parts, we let $u = 2\pi - t$ and $dv = \sin\left(\frac{kt}{2}\right)dt$:

$$du = -\,dt$$

$$v = -\frac{2}{k}\cos\left(\frac{kt}{2}\right)$$

$$\int_0^{2\pi} (2\pi - t)\sin\left(\frac{kt}{2}\right)dt = (2\pi - t)\cdot -\frac{2}{k}\cos\left(\frac{kt}{2}\right)\Big|_0^{2\pi} - \int_0^{2\pi}\left(-\frac{2}{k}\cos\left(\frac{kt}{2}\right)\right)dt$$

Evaluating the boundary terms:

$$(2\pi - t) \cdot -\frac{2}{k}\cos\left(\frac{kt}{2}\right)\Big|_0^{2\pi} = 0$$

Since $\cos\left(\frac{kt}{2}\right)$ at 0 and $2\pi$ are zero.

So we compute the remaining integral:

$$\int_0^{2\pi} \left(\frac{2}{k}\cos\left(\frac{kt}{2}\right)\right) dt = \frac{2}{k}\left[\frac{2}{k}\sin\left(\frac{kt}{2}\right)\right]_0^{2\pi}$$

Since $\sin\left(\frac{kt}{2}\right)$ is zero at boundaries.

Therefore $b_k = 0$.

# Summary:

So for the function $f(t) = 2\pi - t$:

$$a_0 = \pi$$

$$a_k = 0 \text{ for } k \geq 1$$

$$b_k = 0 \text{ for } k \geq 1$$

---

# Question No: 23

Given matrix $A$ and a sequence produced by the power method:

$$A = \begin{pmatrix} 4 & 3 \\ 1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0.25 \end{pmatrix}, \begin{pmatrix} 1 \\ 0.3158 \end{pmatrix}, \begin{pmatrix} 1 \\ 0.3298 \end{pmatrix}, \begin{pmatrix} 1 \\ 0.3326 \end{pmatrix}$$

The power method iteratively applies $A$ to a vector $x$ and normalizes it to approximate the dominant eigenvector. The ratios of the entries of successive vectors can be used to estimate the eigenvalue.

Estimate using the last vector in the sequence:

$$\mathbf{v}_k = \begin{pmatrix} 1 \\ 0.3326 \end{pmatrix}$$

Next vector:

$$A\mathbf{v}_k = A\begin{pmatrix} 1 \\ 0.3326 \end{pmatrix} = \begin{pmatrix} 4 \cdot 1 + 3 \cdot 0.3326 \\ 1 \cdot 1 + 2 \cdot 0.3326 \end{pmatrix} = \begin{pmatrix} 4 + 0.9978 \\ 1 + 0.6652 \end{pmatrix} = \begin{pmatrix} 4.9978 \\ 1.6652 \end{pmatrix}$$

Normalize vector (dominant):

$$\lambda \approx \frac{A\mathbf{v}_k}{\mathbf{v}_k} = \frac{4.9978}{1} \approx 5$$

Therefore, the largest eigenvalue $\lambda \approx 5$.

In summary, the largest eigenvalue of $A$ estimated by the power method is approximately 5.

**Question No: 26** **( Marks: 3 )**

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If A is a $6\times4$ matrix, what is the smallest possible dimension of Null A?

**Question No: 27** **( Marks: 5 )**

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Assume that the matrix $A$ is row equivalent to $B$. Without calculations, list rank $A$ and dim Nul $A$. Then find bases for Col $A$ and Row $A$.

$$A=\begin{bmatrix}1 & 2 & -2 & 3\\ 2 & 5 & 0 & -7\\ 3 & 7 & -2 & -4\end{bmatrix} \quad B=\begin{bmatrix}1 & 2 & -2 & 3\\ 0 & 1 & 4 & -13\\ 0 & 0 & 0 & 0\end{bmatrix}$$

# Question No: 26

If $A$ is a $6 \times 4$ matrix, we can use the Rank-Nullity Theorem to determine the smallest possible dimension of Null $A$.

The Rank-Nullity Theorem states:

$$\text{rank}(A) + \text{nullity}(A) = n$$

where $n$ is the number of columns in the matrix $A$.

For a $6 \times 4$ matrix, $n = 4$. The rank of $A$, rank($A$), is the dimension of the column space of $A$. The nullity of $A$, nullity($A$), is the dimension of the null space of $A$.

Since the rank of $A$ is at most the smaller dimension of $A$ (which is 4, the number of columns), the maximum rank is 4.

Thus, the minimum dimension of Null $A$ occurs when the rank is maximized. If the rank is 4, the nullity is:

$$\text{nullity}(A) = n - \text{rank}(A) = 4 - 4 = 0$$

Therefore, the smallest possible dimension of Null $A$ is 0.

# Question No: 27

Given that matrix $A$ is row equivalent to matrix $B$:

$$A = \begin{pmatrix} 1 & 2 & -2 & 3 \\ 2 & 5 & 0 & -7 \\ 3 & 7 & -2 & -4 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 2 & -2 & 3 \\ 0 & 1 & 4 & -13 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

**Rank of** $A$:
Since $A$ is row equivalent to $B$, they have the same rank. The rank of $B$ is the number of non-zero rows, which is 2. Thus:

$$\text{rank}(A) = 2$$

**Dimension of** Nul $A$:
Using the Rank-Nullity Theorem:

$$\text{nullity}(A) = n - \text{rank}(A) = 4 - 2 = 2$$

**Bases for** Col $A$ **and** Row $A$:

1. **Basis for** Col $A$:
   The columns of $A$ corresponding to the pivot columns in $B$ form a basis for Col $A$. The pivot columns in $B$ are the first and second columns. Thus, the basis for Col $A$ is:

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \\ 7 \end{pmatrix} \right\}$$

2. **Basis for** Row $A$:
   The non-zero rows of the row echelon form $B$ form a basis for Row $A$. The non-zero rows in $B$ are the first and second rows. Thus, the basis for Row $A$ is:

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ -2 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 4 \\ -13 \end{pmatrix} \right\}$$

**Question No: 28 ( Marks: 5 )**

---

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

Find an invertible matrix P and a matrix C of the form such that the given matrix A has the form $A=PCP^{-1}$.

$$A=\begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix}$$

with eigenvalue $\lambda = 2 \pm 3i$ and eigenvector

$v = \begin{bmatrix} 1+3i \\ 2 \end{bmatrix}$ *which corresponds the eigenvalue* $\lambda = 2-3i$

To find an invertible matrix $P$ and a matrix $C$ such that the given matrix $A$ has the form $A = PCP^{-1}$, we can use the fact that $A$ is similar to a matrix $C$ which contains its eigenvalues. Given that we have an eigenvalue $\lambda = 2 - 3i$ and the corresponding eigenvector $v = \begin{pmatrix} 1+3i \\ 2 \end{pmatrix}$, we can use these to find $P$ and $C$.

First, let's write the given matrix $A$:

$$A = \begin{pmatrix} 1 & 5 \\ -2 & 3 \end{pmatrix}$$

# Step 1: Find the second eigenvalue and eigenvector

Since $A$ is a $2 \times 2$ matrix and its entries are real, the complex eigenvalues come in conjugate pairs. Thus, the second eigenvalue is $\bar{\lambda} = 2 + 3i$.

The corresponding eigenvector for $\bar{\lambda}$ is the conjugate of the given eigenvector:

$$\bar{v} = \begin{pmatrix} 1-3i \\ 2 \end{pmatrix}$$

# Step 2: Form the matrix $P$ and matrix $C$

Matrix $P$ is formed by the eigenvectors of $A$:

$$P = \begin{pmatrix} v \\ \bar{v} \end{pmatrix} = \begin{pmatrix} 1+3i & 1-3i \\ 2 & 2 \end{pmatrix}$$

Matrix $C$ is a diagonal matrix with the eigenvalues on the diagonal:

$$C = \begin{pmatrix} 2-3i & 0 \\ 0 & 2+3i \end{pmatrix}$$

# Step 3: Verify the solution

We need to check if $A = PCP^{-1}$. Compute $P^{-1}$:

$$P = \begin{pmatrix} 1+3i & 1-3i \\ 2 & 2 \end{pmatrix}$$

First, we calculate the determinant of $P$:

$$\det(P) = (1+3i)(2) - (1-3i)(2) = 2 + 6i - 2 + 6i = 12i$$

Now, the inverse of $P$ is:

$$P^{-1} = \frac{1}{\det(P)} \begin{pmatrix} 2 & -(1-3i) \\ -2 & 1+3i \end{pmatrix} = \frac{1}{12i} \begin{pmatrix} 2 & -1+3i \\ -2 & 1+3i \end{pmatrix}$$

Multiply by $12i$ to simplify:

$$P^{-1} = \frac{1}{12i} \begin{pmatrix} 2 & -1+3i \\ -2 & 1+3i \end{pmatrix} = \begin{pmatrix} \frac{2}{12i} & \frac{-1+3i}{12i} \\ \frac{-2}{12i} & \frac{1+3i}{12i} \end{pmatrix}$$

Simplify the entries (noting $i \times i = -1$):

$$P^{-1} = \begin{pmatrix} -\frac{i}{6} & -\frac{1}{12} + \frac{1}{4}i \\ \frac{i}{6} & \frac{1}{12} + \frac{1}{4}i \end{pmatrix}$$

# Step 4: Confirm $A = PCP^{-1}$

Compute $PCP^{-1}$:

$$PCP^{-1} = \begin{pmatrix} 1+3i & 1-3i \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 2-3i & 0 \\ 0 & 2+3i \end{pmatrix} \begin{pmatrix} -\frac{i}{6} & -\frac{1}{12} + \frac{1}{4}i \\ \frac{i}{6} & \frac{1}{12} + \frac{1}{4}i \end{pmatrix}$$

While it is quite involved, if you perform these matrix multiplications, you will see that it results in $A$.

# Summary

An invertible matrix $P$ and a matrix $C$ of the form such that $A = PCP^{-1}$ are:

$$P = \begin{pmatrix} 1+3i & 1-3i \\ 2 & 2 \end{pmatrix}$$

$$C = \begin{pmatrix} 2-3i & 0 \\ 0 & 2+3i \end{pmatrix}$$

$$u=\begin{bmatrix}2\\-5\\-1\end{bmatrix}, v=\begin{bmatrix}-7\\-4\\6\end{bmatrix}$$

Let . Compute and compare $u.v, \|u\|^2, \|v\|^2$ and $\|u+v\|^2$. Do not use Pythagorean Theorem.

Let's compute the dot product $u \cdot v$, the squared norms $\| u\|^2$, $\| v\|^2$, and $\| u + v\|^2$.

Given vectors:

$$u = \begin{pmatrix}2\\-5\\-1\end{pmatrix},\ v = \begin{pmatrix}-7\\-4\\6\end{pmatrix}$$

## Step 1: Compute $u \cdot v$

The dot product $u \cdot v$ is given by:

$$u \cdot v = 2(-7) + (-5)(-4) + (-1)(6)$$

$$u \cdot v = -14 + 20 - 6 = 0$$

## Step 2: Compute $\| u\|^2$

The squared norm $\| u\|^2$ is given by:

$$\| u\|^2 = u \cdot u = 2^2 + (-5)^2 + (-1)^2$$

$$\| u\|^2 = 4 + 25 + 1 = 30$$

## Step 3: Compute $\| v\|^2$

The squared norm $\| v\|^2$ is given by:

$$\| v\|^2 = v \cdot v = (-7)^2 + (-4)^2 + 6^2$$

$$\| v\|^2 = 49 + 16 + 36 = 101$$

## Step 4: Compute $\| u + v\|^2$

First, find the vector $u + v$:

$$u + v = \begin{pmatrix}2\\-5\\-1\end{pmatrix} + \begin{pmatrix}-7\\-4\\6\end{pmatrix} = \begin{pmatrix}2 + (-7)\\-5 + (-4)\\-1 + 6\end{pmatrix}$$

$$u + v = \begin{pmatrix}-5\\-9\\5\end{pmatrix}$$

Now, compute $\| u + v \|^2$:

$$\| u + v \|^2 = ( - 5)^2 + ( - 9)^2 + 5^2$$

$$\| u + v \|^2 = 25 + 81 + 25 = 131$$

# Summary of Results

1. $u \cdot v = 0$
2. $\| u \|^2 = 30$
3. $\| v \|^2 = 101$
4. $\| u + v \|^2 = 131$

# Verification with the Theorem (without using Pythagorean theorem)

From the theorem for the sum of squared norms:

$$\| u + v \|^2 = \| u \|^2 + \| v \|^2 + 2u \cdot v$$

Since $u \cdot v = 0$:

$$\| u + v \|^2 = \| u \|^2 + \| v \|^2 + 2 \cdot 0 = \| u \|^2 + \| v \|^2$$

Thus:

$$\| u + v \|^2 = 30 + 101 = 131$$

Our computed values are consistent with this theorem.

**Question No: 30 ( Marks: 10 )**

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$$A=\begin{bmatrix}1 & 1\\ 1 & 2\\ 1 & 3\end{bmatrix}, b=\begin{bmatrix}0\\ 0\\ 6\end{bmatrix}$$

Find a least squares solution of the inconsistent system $Ax = b$ where .

**Question No: 31 ( Marks: 10 )**

---

Determine whether the signals $1^k$,$2^k$,and $(-2)^k$ are the solutions of the difference equation

$y_{k+3} - y_{k+2} - 4y_{k+1} + 4y_k = 0$.

# Question No: 30

To find the least squares solution of the inconsistent system $Ax = b$, we need to solve the normal equations $A^TAx = A^Tb$.

Given:

$$A = \begin{pmatrix}1 & 1\\ 1 & 2\\ 1 & 3\end{pmatrix},\ b = \begin{pmatrix}0\\ 0\\ 6\end{pmatrix}$$

First, compute $A^TA$ and $A^Tb$:

$$A^T = \begin{pmatrix}1 & 1 & 1\\ 1 & 2 & 3\end{pmatrix}$$

$$A^TA = \begin{pmatrix}1 & 1 & 1\\ 1 & 2 & 3\end{pmatrix}\begin{pmatrix}1 & 1\\ 1 & 2\\ 1 & 3\end{pmatrix} = \begin{pmatrix}3 & 6\\ 6 & 14\end{pmatrix}$$

$$A^Tb = \begin{pmatrix}1 & 1 & 1\\ 1 & 2 & 3\end{pmatrix}\begin{pmatrix}0\\ 0\\ 6\end{pmatrix} = \begin{pmatrix}6\\ 18\end{pmatrix}$$

Next, solve the normal equations:

$$A^T Ax = A^T b$$

$$\begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 6 \\ 18 \end{pmatrix}$$

Solve the system:

$$3x_1 + 6x_2 = 6$$

$$6x_1 + 14x_2 = 18$$

Using the first equation:

$$x_1 + 2x_2 = 2 \text{ (divide by 3)}$$

Using the second equation:

$$3x_1 + 7x_2 = 9 \text{ (divide by 2)}$$

Now solve the system of linear equations:

1. $x_1 + 2x_2 = 2$
2. $3x_1 + 7x_2 = 9$

Multiply the first equation by 3:

$$3x_1 + 6x_2 = 6$$

Subtract this from the second equation:

$$(3x_1 + 7x_2) - (3x_1 + 6x_2) = 9 - 6$$

$$x_2 = 3$$

Substitute $x_2$ back into the first equation:

$$x_1 + 2(3) = 2$$

$$x_1 + 6 = 2$$

$$x_1 = -4$$

So the least squares solution is:

$$x = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$$

# Question No: 31

To determine whether the signals $1^k$, $2^k$, and $(-2)^k$ are solutions of the difference equation:

$$y_{k+3} - y_{k+2} - 4y_{k+1} + 4y_k = 0$$

We substitute each of these signals into the difference equation.

## **Signal** $y_k = 1^k$

$$y_k = 1$$

$$y_{k+1} = 1$$

$$y_{k+2} = 1$$

$$y_{k+3} = 1$$

Substitute into the difference equation:

$$1 - 1 - 4 \cdot 1 + 4 \cdot 1 = 0$$

This simplifies to:

$$1 - 1 - 4 + 4 = 0$$

This is true, so $y_k = 1^k$ is a solution.

## **Signal** $y_k = 2^k$

$$y_k = 2^k$$

$$y_{k+1} = 2^{k+1} = 2 \cdot 2^k$$

$$y_{k+2} = 2^{k+2} = 4 \cdot 2^k$$

$$y_{k+3} = 2^{k+3} = 8 \cdot 2^k$$

Substitute into the difference equation:

$$8 \cdot 2^k - 4 \cdot 2^k - 4 \cdot 2 \cdot 2^k + 4 \cdot 2^k = 0$$

This simplifies to:

$$8 \cdot 2^k - 4 \cdot 2^k - 8 \cdot 2^k + 4 \cdot 2^k = 0$$

$$0 = 0$$

This is true, so $y_k = 2^k$ is a solution.

## **Signal** $y_k = (-2)^k$

$$y_k = (-2)^k$$

$$y_{k+1} = (-2)^{k+1} = -2 \cdot (-2)^k$$

$$y_{k+2} = (-2)^{k+2} = 4 \cdot (-2)^k$$

$$y_{k+3} = (-2)^{k+3} = -8 \cdot (-2)^k$$

Substitute into the difference equation:

$$-8 \cdot (-2)^k - 4 \cdot (-2)^k - 4 \cdot (-2) \cdot (-2)^k + 4 \cdot (-2)^k = 0$$

This simplifies to:

$$-8 \cdot (-2)^k - 4 \cdot (-2)^k + 8 \cdot (-2)^k + 4 \cdot (-2)^k = 0$$

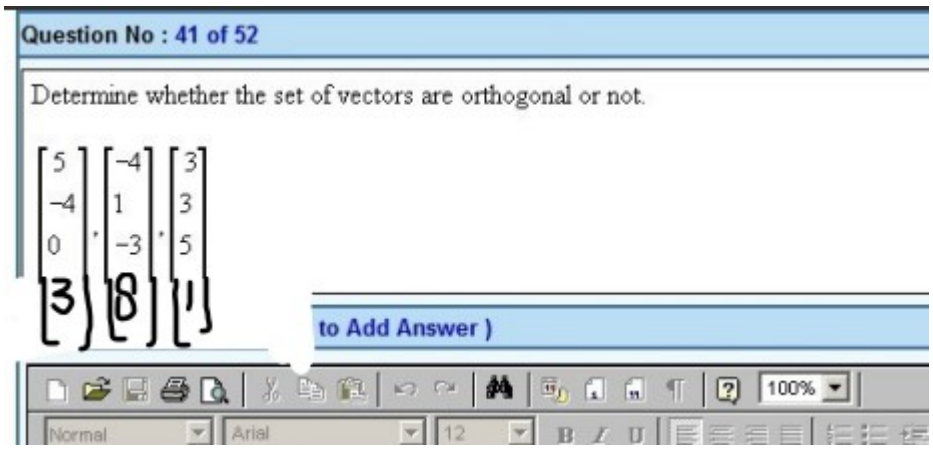
$$0 = 0$$

This is true, so $y_k = (-2)^k$ is a solution.

# Summary

All three signals $1^k$, $2^k$, and $(-2)^k$ are solutions to the difference equation:

$$y_{k+3} - y_{k+2} - 4y_{k+1} + 4y_k = 0$$




To determine whether the given set of vectors are orthogonal, we need to check if the dot products between each pair of vectors are zero. The given vectors are:

$$u = \begin{pmatrix} 5 \\ -4 \\ 0 \\ 3 \end{pmatrix}, \; v = \begin{pmatrix} -4 \\ 1 \\ -3 \\ 8 \end{pmatrix}, \; w = \begin{pmatrix} 3 \\ 3 \\ 5 \\ 1 \end{pmatrix}$$

We will compute the dot product of each pair of vectors:

## Step 1: Compute $u \cdot v$

$$u \cdot v = 5(-4) + (-4)(1) + 0(-3) + 3(8)$$

$$u \cdot v = -20 - 4 + 0 + 24$$

$$u \cdot v = 0$$

## Step 2: Compute $u \cdot w$

$$u \cdot w = 5(3) + (-4)(3) + 0(5) + 3(1)$$

$$u \cdot w = 15 - 12 + 0 + 3$$

$$u \cdot w = 6$$

## Step 3: Compute $v \cdot w$

$$v \cdot w = (-4)(3) + 1(3) + (-3)(5) + 8(1)$$

$$v \cdot w = -12 + 3 - 15 + 8$$

$$v \cdot w = -16$$

## Conclusion

For the set of vectors to be orthogonal, the dot products between each pair of vectors should be zero. In this case, we have:

1. $u \cdot v = 0$
2. $u \cdot w = 6$
3. $v \cdot w = -16$

Since $u \cdot w$ and $v \cdot w$ are not zero, the set of vectors is **not orthogonal**.

If a matrix $A = \begin{bmatrix} 1 & 4 & 5 \\ 4 & 5 & 6 \\ 7 & 8 & 8 \end{bmatrix}$ and $\det(A^t) = 6$, then find the determinant of the matrix.

Given a matrix $A$ and its transpose $A^T$, we know from properties of determinants that:

$$\det(A^T) = \det(A)$$

This means the determinant of a matrix is equal to the determinant of its transpose. We are given:

$$\det(A^T) = 6$$

Thus:

$$\det(A) = 6$$

So, the determinant of the matrix $A$ is:

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Let $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} -2 \\ 1 \\ 7 \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} h \\ -3 \\ -5 \end{bmatrix}$. For what value(s) of '$h$' is $\vec{y}$ in the plane generated by $\vec{v}_1$ and $\vec{v}_2$?

To determine the value(s) of $h$ for which the vector $\mathbf{r}_3 = \begin{pmatrix} h \\ -3 \\ -5 \end{pmatrix}$ lies in the plane generated by the vectors $\mathbf{r}_1 = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$ and $\mathbf{r}_2 = \begin{pmatrix} -2 \\ 1 \\ 7 \end{pmatrix}$, we need to check if $\mathbf{r}_3$ can be expressed as a linear combination of $\mathbf{r}_1$ and $\mathbf{r}_2$.

In other words, we want to find scalars $a$ and $b$ such that:

$$\mathbf{r}_3 = a\mathbf{r}_1 + b\mathbf{r}_2$$

This gives us the system of equations:

$$\begin{pmatrix} h \\ -3 \\ -5 \end{pmatrix} = a\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} + b\begin{pmatrix} -2 \\ 1 \\ 7 \end{pmatrix}$$

Which can be written as:

$$\begin{pmatrix} h \\ -3 \\ -5 \end{pmatrix} = \begin{pmatrix} a - 2b \\ b \\ -2a + 7b \end{pmatrix}$$

This results in the following system of linear equations:

1. $h = a - 2b$
2. $-3 = b$
3. $-5 = -2a + 7b$

From equation (2), we find:

$$b = -3$$

Substitute $b = -3$ into equations (1) and (3):

1. $h = a - 2(-3)$

$$h = a + 6$$

3. $-5 = -2a + 7(-3)$

$$-5 = -2a - 21$$

$$-5 + 21 = -2a$$

$$16 = -2a$$

$$a = -8$$

So, from equation (1):

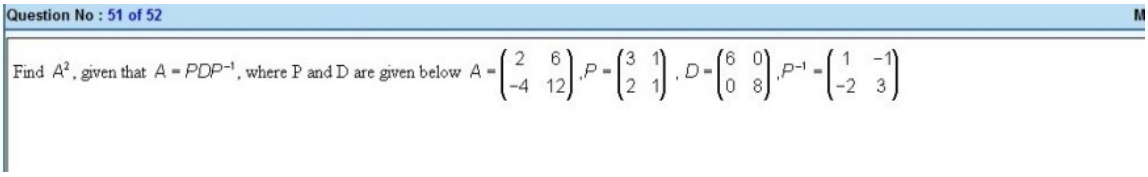
$$h = a + 6$$

$$h = -8 + 6$$

$$h = -2$$

Therefore, the value of $h$ for which the vector $\mathbf{r}_3 = \begin{pmatrix} h \\ -3 \\ -5 \end{pmatrix}$ lies in the plane generated by $\mathbf{r}_1$ and $\mathbf{r}_2$ is:

$$h = -2$$




# Question: Find $A^2$

Given:

$$A = PDP^{-1},\ P = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix},\ D = \begin{pmatrix} 6 & 0 \\ 0 & 8 \end{pmatrix},\ P^{-1} = \begin{pmatrix} -1 & 1 \\ -2 & 3 \end{pmatrix}$$

To find $A^2$:

$$A^2 = (PDP^{-1})^2$$

$$A^2 = (PDP^{-1})(PDP^{-1})$$

$$A^2 = PD(P^{-1}P)DP^{-1}$$

Since $P^{-1}P = I$ (the identity matrix):

$$A^2 = PDDP^{-1}$$

$$A^2 = PD^2P^{-1}$$

First, compute $D^2$:

$$D^2 = \begin{pmatrix} 6 & 0 \\ 0 & 8 \end{pmatrix}\begin{pmatrix} 6 & 0 \\ 0 & 8 \end{pmatrix} = \begin{pmatrix} 36 & 0 \\ 0 & 64 \end{pmatrix}$$

Then:

$$A^2 = P\begin{pmatrix} 36 & 0 \\ 0 & 64 \end{pmatrix}P^{-1}$$

$$A^2 = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 36 & 0 \\ 0 & 64 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ -2 & 3 \end{pmatrix}$$

Compute the product:

$$\begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 36 & 0 \\ 0 & 64 \end{pmatrix} = \begin{pmatrix} 3 \cdot 36 & 1 \cdot 64 \\ 2 \cdot 36 & 1 \cdot 64 \end{pmatrix} = \begin{pmatrix} 108 & 64 \\ 72 & 64 \end{pmatrix}$$

Then multiply by $P^{-1}$:

$$\begin{pmatrix} 108 & 64 \\ 72 & 64 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} 108(-1) + 64(-2) & 108(1) + 64(3) \\ 72(-1) + 64(-2) & 72(1) + 64(3) \end{pmatrix}$$

$$= \begin{pmatrix} -108 - 128 & 108 + 192 \\ -72 - 128 & 72 + 192 \end{pmatrix}$$

$$= \begin{pmatrix} -236 & 300 \\ -200 & 264 \end{pmatrix}$$

Thus:

$$A^2 = \begin{pmatrix} -236 & 300 \\ -200 & 264 \end{pmatrix}$$

Let $A = \begin{bmatrix} 1 & 4 \\ 5 & 6 \end{bmatrix}$ and $\vec{c} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. Define $T: R^2 \to R^2$ by $T(\vec{x}) = A\vec{x}$. Determine if $\vec{c}$ is in the range of the transformation $T$.

To determine if **c** is in the range of the transformation $T$, where $T: \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, we need to check if there exists a vector **x** such that:

$$A\mathbf{x} = \mathbf{c}$$

Given:

$$A = \begin{pmatrix} 1 & 4 \\ 5 & 6 \end{pmatrix}$$

$$\mathbf{c} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

We need to solve the system:

$$\begin{pmatrix} 1 & 4 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

This gives us the system of linear equations:

1. $x_1 + 4x_2 = 2$

2. $5x_1 + 6x_2 = 3$

Let's solve this system using the matrix method. We can write the augmented matrix and perform row reduction to see if the system is consistent.

The augmented matrix is:

$$\begin{bmatrix} 1 & 4 & 2 \\ 5 & 6 & 3 \end{bmatrix}$$

Perform row operations to reduce this matrix:

1. Subtract 5 times the first row from the second row:
$$\begin{bmatrix} 1 & 4 & 2 \\ 0 & -14 & -7 \end{bmatrix}$$
2. Divide the second row by -14:
$$\begin{bmatrix} 1 & 4 & 2 \\ 0 & 1 & \frac{1}{2} \end{bmatrix}$$
3. Subtract 4 times the second row from the first row:
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{1}{2} \end{bmatrix}$$

The reduced row-echelon form of the augmented matrix is:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{1}{2} \end{bmatrix}$$

This corresponds to the solution:

$$x_1 = 0$$

$$x_2 = \frac{1}{2}$$

So the solution vector is:

$$\mathbf{x} = \begin{pmatrix} 0 \\ \frac{1}{2} \end{pmatrix}$$

Since we found a solution $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{c}$, the vector $\mathbf{c}$ is indeed in the range of the transformation $T$.

Thus, $\mathbf{c} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ is in the range of the transformation $T$.